Maple Thermodynamics Academy

Unit 1 learning workbook

Six complete chapters, worked examples, original diagrams, formula packet, glossary and review sheet.

All competition levels | Student edition | Batch 1

Study in order: quantities, systems, property types, equilibrium, temperature scales, energy units. Use the separate practice packet for WS01–WS06 and D01.

The complete seven-unit curriculum is in production; this workbook contains Unit 1 only.

U1-C01 - Heat, temperature, thermal energy

Unit 1 | All competition levels | public-student

Learning objectives

Distinguish temperature, internal energy and heat; predict the direction of spontaneous heat transfer; explain why sample size matters.

Prerequisites

Compare numbers; recognize grams and degrees Celsius. No formula knowledge is needed.

Learn the idea

Imagine a teaspoon of warm water and a bathtub of water at the same temperature. A thermometer can show the same reading in both. Yet the bathtub can transfer much more energy as it cools. Temperature tells us something about the thermal state, not how much material is present. For samples of the same material in the same phase, at the same temperature and under comparable conditions, a larger sample has a larger total internal energy.

Matter consists of particles. Even a solid's particles move: they vibrate around relatively fixed positions. In a simple ideal-gas model, temperature is linked to average random translational kinetic energy. For liquids and solids, particles also interact strongly, so 'temperature is average particle speed' is too crude. Different particle masses can also mean different typical speeds at the same temperature.

Internal energy is the microscopic energy associated with the system's particles and their interactions. It excludes the motion of the entire object across a room and its gravitational energy as a whole. Thermal energy is a useful classroom term for the temperature-related part of microscopic energy; sources use the phrase somewhat differently. In this course, use internal energy U when making an energy balance. During a phase change, internal energy can change without a temperature change because particle interactions change.

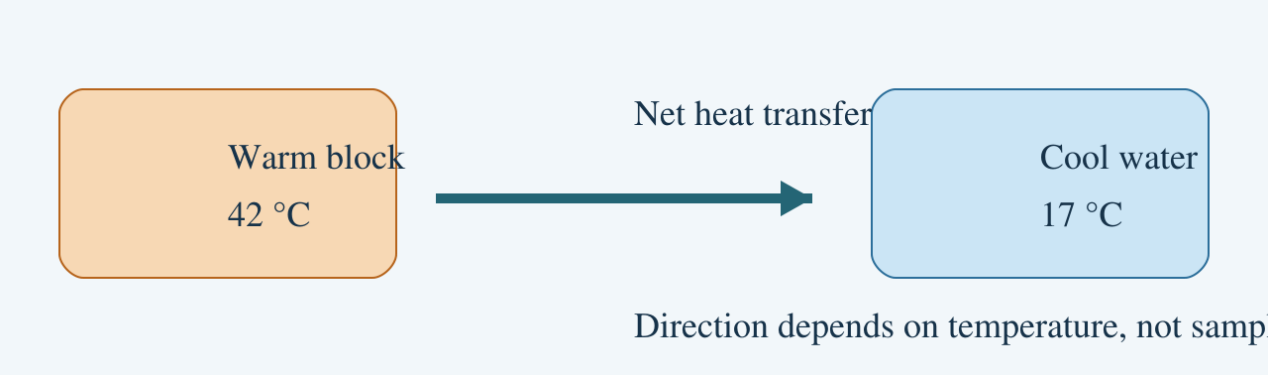
Heat is a transfer, not a stored material. Put a warm metal spoon into cooler water: energy transfers from spoon to water because their temperatures differ. We call that energy transfer heat. We say 'the water gained internal energy by heating,' rather than 'the water contains heat.' If their temperatures match, there is no net heat transfer between them, although particles still move and microscopic exchanges continue.

A useful boundary sketch is: warmer spoon | boundary | cooler water. Label an energy arrow from the spoon toward the water. If the water is your system, the energy crosses into your system. Turning the drawing around does not change the physical transfer; it only changes which object you chose to study.

In the introductory energy examples below, assume no work and no other transfers. Then energy gained by the cool object equals energy lost by the warm object. In later units we will include work and environmental losses. These assumptions are part of the explanation, not invisible

conveniences.

Visual model



Definitions and units

Temperature T: thermal-state reading, in °C or K. Internal energy U: microscopic system energy, in J. Heat Q: energy transferred because of a temperature difference, in J. Net transfer: transfer in one direction minus transfer in the opposite direction.

Equations and their meaning

For an isolated two-object exchange with no work: energy lost by one = energy gained by the other. Energy units measure transfers; temperature units measure a different kind of quantity.

Worked example 1: Energy received by water

Given: The spoon transfers 180 J to the water; there is no other transfer or work.

Find: Increase in the water’s internal energy.

Equation: Energy gain = energy received.

Substitution: Energy gain = 180 J.

Arithmetic: No conversion is needed.

Answer with units: The water’s internal energy increases by 180 J.

Physical interpretation: This is not an increase of 180 °C; temperature change also depends on mass and material.

Worked example 2: An energy ledger

Given: A sample receives 320 J and later transfers 90 J out; no work occurs.

Find: Net internal-energy change.

Equation: Net change = energy in − energy out.

Substitution: Net change = 320 J − 90 J.

Arithmetic: 320 − 90 = 230.

Answer with units: The increase is 230 J.

Physical interpretation: Energy can enter and leave during one interval. Count the net change.

Common confusions

A hotter small sample need not have more total internal energy than a cooler large one. A large internal energy does not determine heat-flow direction; temperature difference does. Equal temperatures do not mean particles stop. Equal temperatures also do not guarantee equal internal energies.

Real-life connection

A hot drop may cool quickly while a warm bath stays warm longer. That difference involves quantity of material and transfer conditions. It does not mean the bath must have the higher temperature.

Recap

Temperature describes thermal state; internal energy belongs to a system; heat crosses a boundary because temperatures differ. State the system and the assumptions before counting energy.

Review questions

1. A 60 °C spoon touches 20 °C water. Which way is the initial net energy transfer?

2. Why can a bucket and a cup at the same temperature have different internal energies?

3. Rewrite “this cup contains 200 J of heat” in a scientifically meaningful way.

Practice

Open WS01. Worksheet questions are separate from the review questions above.

Mastery checklist

☐ I can label heat as a transfer in a boundary sketch.

☐ I can explain why temperature is not a total-energy meter.

☐ I can compute an energy ledger without mixing J and °C.

U1-C02 – Thermodynamic systems and surroundings

Unit 1 | All competition levels | public-student

Learning objectives

Choose a useful system boundary; classify open, closed and ideal isolated systems; distinguish matter exchange from energy exchange.

Prerequisites

Chapter U1-C01; understand “inside,” “outside,” and transfer.

Learn the idea

A system is the matter or region we choose to study. Everything outside it is the surroundings. The boundary separates the two. It can be an actual wall or an imaginary surface. There is no single mandatory system for a physical situation: you might study the water in a cup, the cup plus water, or the whole room. A good choice makes the question easier to answer.

A boundary is not necessarily an energy barrier. A sealed bottle can exchange energy through its walls while keeping the same matter inside. Similarly, an open cup can exchange energy and lose water molecules by evaporation. Ask two separate questions: Can matter cross? Can energy cross?

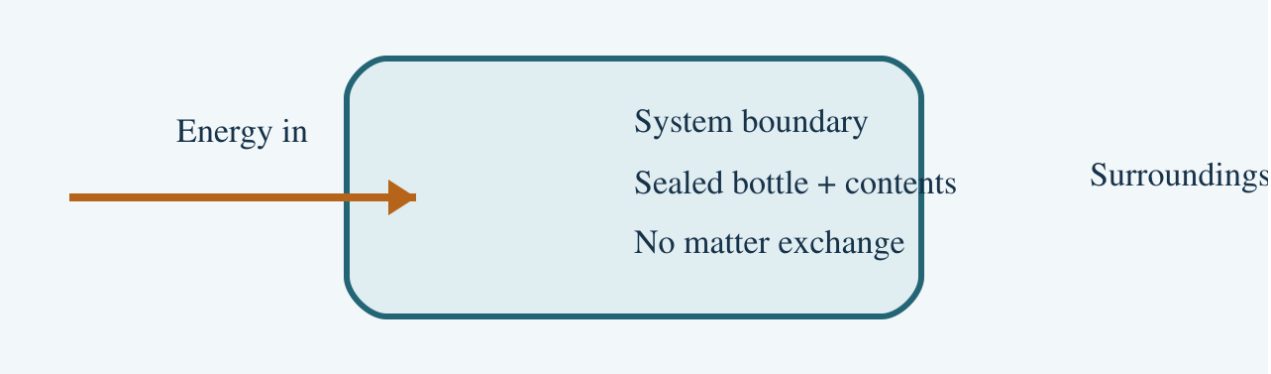
An open system can exchange both matter and energy with its surroundings. A closed system can exchange energy but not matter. An isolated system exchanges neither. “Closed” does not mean cold, insulated or motionless. A closed gas cylinder with a movable piston can exchange energy by work as its boundary moves while retaining all its gas.

Real insulation slows transfer; it does not automatically make a system perfectly isolated. A sealed, well-insulated container can be modeled as approximately isolated for a short interval if the neglected energy exchange is small enough for the task. Always identify this as a model. Do not claim that an ordinary insulated container guarantees zero transfer for all time.

Draw a rectangle labeled “water only.” Put “air” and “cup” outside it. A vapor arrow crossing out of the rectangle means matter leaves, making this an open system. Now draw a different rectangle around a sealed bottle and its contents. With no leaks, no matter crosses that larger boundary, but an energy arrow can still cross from warmer air. That is a closed system. The classification follows the stated boundary and conditions, not the everyday word “container.”

System bookkeeping also works for mass. If water enters and water leaves, the final mass equals the starting mass plus inflow minus outflow. Use a single unit before subtracting. For energy, a transfer across the boundary changes the system's energy unless another transfer offsets it. Later, we separate heat and work; for now, name the physical route in words.

Visual model



Definitions and units

System: selected matter or region. Surroundings: everything outside. Boundary: separating surface. Open: matter and energy exchange allowed. Closed: energy exchange allowed, matter exchange excluded. Isolated: neither exchange occurs in the model.

Equations and their meaning

Mass bookkeeping: m_final = m_start + m_in − m_out. In a no-work energy example: net energy gain = energy in − energy out. All mass terms must share units; all energy terms must share units.

Worked example 1: Mass through an open boundary

Given: An open vessel initially holds 150 g water; 25 g is poured in and 8 g evaporates.

Find: Final water mass.

Equation: m_final = m_start + m_in − m_out.

Substitution: m_final = 150 g + 25 g − 8 g.

Arithmetic: 150 + 25 − 8 = 167.

Answer with units: 167 g.

Physical interpretation: Matter crossed the boundary in both directions, so the water system is open.

Worked example 2: A sealed bottle can warm

Given: A sealed bottle and its contents receive 85 J through the walls; 20 J later leaves, with no work.

Find: Net energy gained.

Equation: Net gain = energy in − energy out.

Substitution: Net gain = 85 J − 20 J.

Arithmetic: 85 − 20 = 65.

Answer with units: 65 J.

Physical interpretation: No matter crossed, yet energy changed: closed does not mean isolated.

Common confusions

A lid alone does not prove that a system is closed: specify whether it is sealed against matter flow. Insulation alone does not prevent matter exchange. A boundary may move. Energy transfer is not the same as matter transfer, although moving matter can carry energy.

Real-life connection

For a device calculation, defining the system as water only makes its temperature gain central; defining it as water plus cup also includes energy stored in the cup. Those are different models of the same trial.

Recap

Name the boundary first. Classify matter exchange and energy exchange separately. Use an isolated model only when its assumptions are justified.

Review questions

1. Is a sealed bottle in sunlight isolated? Explain.

2. How would your boundary differ for water only versus cup plus water?

3. Can a closed system exchange work?

Practice

Open WS02. Worksheet questions are separate from the review questions above.

Mastery checklist

☐ I can draw and label a system boundary.

☐ I can classify exchange using two separate questions.

☐ I can explain the limitations of an isolated-system approximation.

U1-C03 - Intensive and extensive properties

Unit 1 | All competition levels | public-student

Learning objectives

Classify intensive and extensive properties; apply the split/combine test; use density and mass consistently.

Prerequisites

Chapters U1-C01–02; division and multiplication; basic volume units.

Learn the idea

Suppose you have a uniform sample of water and gently separate it into two equal portions without changing its temperature or pressure appreciably. Each portion has half the original mass and half the original volume. Its temperature is unchanged. This thought experiment distinguishes properties that scale with the amount of material from properties that do not.

An extensive property scales with system size when comparable portions of a homogeneous sample are combined. Mass, volume and total internal energy are extensive. An intensive property does not scale simply with the amount. Temperature, pressure and density are intensive. These labels describe how a property behaves under a controlled comparison; they do not mean that the property can never change.

For example, density can change with temperature or composition, but doubling an otherwise identical uniform sample does not double its density. Pressure can change if you compress a gas, but that does not make pressure extensive. The split/combine thought experiment must preserve the relevant physical conditions. Squeezing twice as much gas into the same volume is not the same experiment as combining two identical portions while allowing volume to double.

Density is mass per volume: $\rho = m/V$. For two identical portions, both numerator and denominator double. Their ratio stays the same. That is why a ratio of two extensive properties can be intensive. Specific heat will later provide another example: heat capacity grows with sample mass, but heat capacity per mass is a material property under stated conditions.

To picture this, draw one rectangle labeled “80 g, 80 mL, 25 °C” and two half-sized rectangles labeled “40 g, 40 mL, 25 °C.” Keep the temperature label unchanged. Do not split the reading into 12.5 °C. This drawing assumes uniform water and negligible exchange during separation.

Internal energy comparisons require care. For identical material in the same phase, temperature and other relevant conditions, adding an identical portion adds its internal energy. But mass alone cannot compare unlike objects at different conditions. We also choose consistent energy reference states when quoting U. For beginner questions, use explicit ratios or stated illustrative energy values rather than assuming that every object's internal energy is a known number.

Heat itself is not a state property of the material. The transferred amount often scales with mass for the same process, but “heat in the cup” should not appear on a list of stored system

properties.

Definitions and units

Extensive property: scales with amount under comparable conditions. Intensive property: does not scale with amount. Homogeneous: uniform composition and properties on the scale considered. Density ρ: mass divided by volume; commonly kg/m³ or g/mL.

Equations and their meaning

ρ=m/V; m=ρV; V=m/ρ. In comparisons at the same state, doubling a uniform sample doubles mass and volume but leaves density unchanged. 1 mL = 1 cm³.

Worked example 1: Calculate an intensive property

Given: A uniform liquid has mass 72 g and volume 80 mL.

Find: Density.

Equation: ρ=m/V.

Substitution: ρ=72 g ÷ 80 mL.

Arithmetic: 72 ÷ 80 = 0.90.

Answer with units: 0.90 g/mL.

Physical interpretation: Each mL has an average mass of 0.90 g under these conditions.

Worked example 2: Combine identical portions

Given: Two portions each have mass 45 g, volume 50 mL and temperature 24 °C.

Find: Combined mass, volume, density and temperature, with no transfer or mixing effects.

Equation: m_total=m_1+m_2; V_total=V_1+V_2; ρ=m_total/V_total.

Substitution: m_total=45+45 g; V_total=50+50 mL; ρ=90/100 g/mL.

Arithmetic: Totals are 90 g and 100 mL; 90/100=0.90.

Answer with units: 90 g; 100 mL; 0.90 g/mL; 24 °C.

Physical interpretation: The extensive quantities doubled. Density and temperature did not.

Common confusions

Intensive does not mean intense, large or fixed forever. Temperature readings do not add when identical samples combine. Total internal energy and temperature are different properties. Density is not the same as mass.

Real-life connection

A small and a large aluminum part can have the same density but very different masses. In device testing, doubling the amount of water changes the thermal task even if its initial temperature is unchanged.

Recap

Use a controlled split/combine comparison. Mass, volume and internal energy scale with amount; temperature, pressure and density do not simply scale with amount.

Review questions

1. Why does density stay constant when an identical sample is doubled?

2. Why is compression a poor test of whether pressure is intensive?

3. Is the energy transferred in a process a stored state property?

Practice

Open WS03. Worksheet questions are separate from the review questions above.

Mastery checklist

☐ I can classify mass, volume, U, temperature, pressure and density.

☐ I can show the density cancellation with numbers.

☐ I can state the assumptions behind the split/combine test.

U1-C04 – Thermal equilibrium and zeroth law

Unit 1 | All competition levels | public-student

Learning objectives

Use thermal equilibrium and the zeroth law; distinguish equilibrium from equal energy; explain how a thermometer obtains a useful reading.

Prerequisites

Heat versus temperature; comparing temperatures; elementary subtraction.

Learn the idea

Put a warm object beside a cooler object with a path for thermal energy transfer. Net heat transfer is initially from the warmer object to the cooler one. Over time their temperatures can approach each other. Thermal equilibrium means there is no net heat transfer between objects in thermal contact because they have the same temperature.

The word net matters. Microscopic motion and exchanges do not stop at equilibrium. Opposing transfers balance statistically. Thermal equilibrium also does not mean that every kind of physical change has ceased; chemical reactions or motion might still matter in another problem. We are discussing the thermal relationship.

The zeroth law makes temperature comparison consistent. If A is in thermal equilibrium with C, and B is also in thermal equilibrium with C, then A and B are in thermal equilibrium with each other. C acts as a common thermal reference. This is why a thermometer can help compare objects without making every pair touch directly.

A thermometer must interact with what it measures. It takes time for its sensing part to approach the sample's temperature. Reading it immediately can give a value between its old temperature and the sample temperature. Ideally the thermometer's effect on the sample is small; a large, very cold probe in a tiny warm sample can change the quantity being measured. A stable reading and appropriate measurement method matter.

Equal temperatures do not require equal masses or equal internal energies. A cup and a large tank of the same water can both be at 23 °C. They can be in thermal equilibrium even though the tank contains much more material. Nor does equal temperature tell you that both must be 0 °C: equilibrium can occur across many temperatures.

A special averaging shortcut is sometimes useful but must be labeled. If two equal masses of the same material in the same phase exchange energy only with each other, have approximately constant equal specific heats, and do no work, their common final temperature is the average of their initial temperatures. This is not a general rule for unequal masses, unlike materials, phase changes, or heat exchange with a container. In Unit 3, calorimetry will explain the shortcut from energy conservation.

An equilibrium diagram can show two objects marked with the same T and two opposing energy arrows of equal size. For a non-equilibrium diagram, use different T labels and a single arrow labeled “net heat transfer” pointing from high T toward low T.

Definitions and units

Thermal equilibrium: equal-temperature condition giving no net heat exchange in thermal contact. Zeroth law: equilibrium through a common reference implies mutual thermal equilibrium. Thermometer response time: time needed for a sensor to approach the measured temperature.

Equations and their meaning

Temperature difference: $\Delta T = T_{high} - T_{low}$. Special equal-sample model only: $T_{final} = (T_1 + T_2)/2$. The zeroth law is a relationship, not a numerical heat formula.

Worked example 1: Use a common reference

Given: A is in equilibrium with a 26.0 °C reference; B is independently in equilibrium with the same unchanged reference.

Find: Temperature difference between A and B.

Equation: $\Delta T = T_A - T_B$.

Substitution: $\Delta T = 26.0\ °C - 26.0\ °C$.

Arithmetic: $26.0 - 26.0 = 0.0$.

Answer with units: 0.0 °C difference.

Physical interpretation: The zeroth law predicts no net heat transfer when A and B are placed in thermal contact.

Worked example 2: Equal-sample model

Given: Equal water masses start at 18.0 °C and 30.0 °C; same constant specific heat, no phase change, no work or environmental exchange, negligible container energy.

Find: Final common temperature.

Equation: $T_{final} = (T_1 + T_2)/2$ for this special model.

Substitution: $T_{final} = (18.0 + 30.0)/2$ °C.

Arithmetic: $48.0/2 = 24.0$.

Answer with units: 24.0 °C.

Physical interpretation: It lies between the starting temperatures. The averaging shortcut is justified here, not for every mixture.

Common confusions

Thermal equilibrium does not mean equal internal energy, no particle motion, or zero temperature. A brief touch does not prove equilibrium has been reached. A maintained temperature difference with continuous energy flow is not mutual thermal equilibrium.

Real-life connection

A classroom thermometer should settle before its value is recorded. In competition, the event supervisor controls the starting and final measurements; students work from those readings rather than adding their own probe.

Recap

Thermal equilibrium is about equal temperature and zero net heat exchange. A common reference connects comparisons. Good measurement considers both response time and disturbance.

Review questions

1. Why can differently sized samples be in thermal equilibrium?
2. What does the zeroth law allow a thermometer to do?
3. When is the arithmetic-mean temperature shortcut valid?

Practice

Open WS04. Worksheet questions are separate from the review questions above.

Mastery checklist

☐ I can apply the three-object zeroth-law argument.

☐ I can explain “no net” without saying “no motion.”

☐ I can list the assumptions for equal-sample averaging.

U1-C05 - Celsius, Fahrenheit, Kelvin

Unit 1 | All competition levels | public-student

Learning objectives

Convert °C, °F and K; distinguish absolute readings from changes; select the appropriate scale for a calculation.

Prerequisites

Signed arithmetic, multiplication/division by 1.8, and one-variable equations.

Learn the idea

Temperature scales attach numbers to a thermal state. They use different zero points and, in the case of Fahrenheit, a different step size. Changing the label does not change the physical sample. A glass of water does not heat up when you rewrite 20 °C as 293.15 K.

Celsius and Kelvin have equally sized steps, but their zeros are offset. The relation is $T_K=T_C+273.15$. Kelvin uses K, without a degree symbol. Absolute zero corresponds to 0 K or −273.15 °C. In the ordinary thermal systems in this course, Kelvin temperatures are nonnegative. Absolute zero is a limiting state; do not describe it simply as all forms of microscopic motion disappearing.

Fahrenheit has smaller steps. A change of 1 °C corresponds to a change of 1.8 °F. The readings also have an offset: $T_F=1.8T_C+32$. To undo that transformation, subtract 32 first and divide by 1.8. Order matters. Dividing first and then subtracting 32 gives a different and incorrect result.

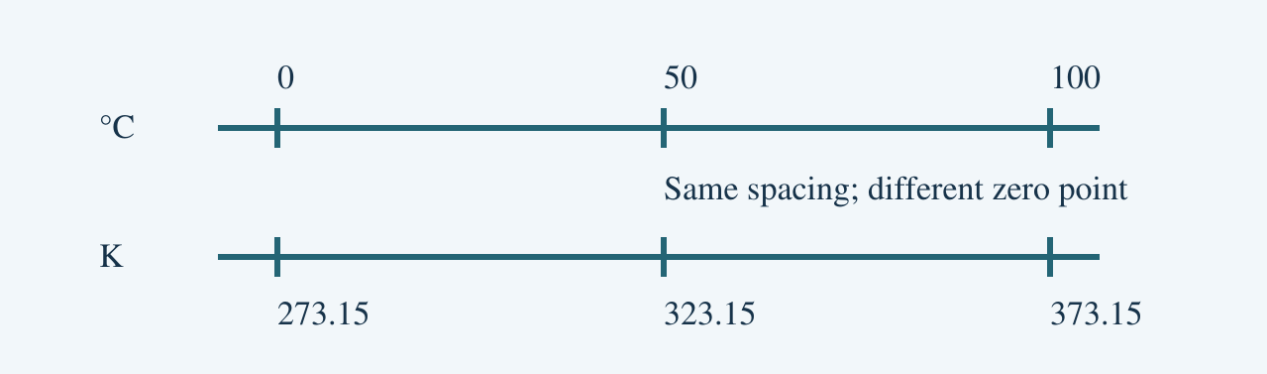
A temperature reading and a temperature change are different quantities. If water goes from 20 °C to 25 °C, its change is +5 °C. In Kelvin it goes from 293.15 K to 298.15 K, a change of +5 K. The offset cancels in the subtraction. Never add 273.15 to a temperature change. For a Fahrenheit change, the 32 also cancels: $\Delta T_F=1.8\Delta T_C$.

Use a number-line sketch with paired points: −273.15 °C ↔ 0 K; 0 °C ↔ 273.15 K; 100 °C ↔ 373.15 K. The last two correspond approximately to water freezing and boiling at standard atmospheric pressure for pure water; pressure affects phase-change temperatures. The 100-degree spacing in Celsius remains a 100-K spacing.

Some equations depend on absolute temperature ratios, such as the ideal gas law introduced later. They require Kelvin. Official Thermodynamics device scoring instead requires Celsius. A unit choice follows the specified equation or rule, not a preference for whichever number looks easier.

For practice, retain digits through intermediate steps and round the final converted reading to a sensible precision. Treat 273.15, 32 and 9/5 as exact scale-conversion constants. A thermometer's measurement precision is not improved just because a conversion calculator prints many decimals.

Visual model



Definitions and units

T_C: Celsius reading in °C. T_F: Fahrenheit reading in °F. T_K: Kelvin reading in K. ΔT: final minus initial temperature, with the scale stated. Absolute temperature: temperature referenced to absolute zero.

Equations and their meaning

T_K=T_C+273.15; T_C=T_K−273.15; T_F=(9/5)T_C+32; T_C=(T_F−32)×5/9. For changes only: ΔT_K=ΔT_C numerically; ΔT_F=(9/5)ΔT_C.

Worked example 1: Room-temperature water in Kelvin

Given: T_C=23.0 °C.

Find: T_K.

Equation: T_K=T_C+273.15.

Substitution: T_K=23.0+273.15 K.

Arithmetic: 23.0+273.15=296.15; round to one decimal.

Answer with units: 296.2 K.

Physical interpretation: This is the same physical temperature, not a warming of the water.

Worked example 2: Convert Fahrenheit to Celsius

Given: T_F=86.0 °F.

Find: T_C.

Equation: T_C=(T_F−32)×5/9.

Substitution: T_C=(86.0−32)×5/9 °C.

Arithmetic: 54.0×5/9=30.0.

Answer with units: 30.0 °C.

Physical interpretation: The reading is warm room-temperature water, consistent with 86 °F.

Worked example 3: A change, not a reading

Given: Water warms from 19.0 °C to 27.0 °C.

Find: Temperature change in K and °F.

Equation: $\Delta T_C = T_{final} - T_{initial}$; $\Delta T_K = \Delta T_C$; $\Delta T_F = 1.8\Delta T_C$.

Substitution: $\Delta T_C = 27.0 - 19.0$; $\Delta T_F = 1.8 \times 8.0$.

Arithmetic: $\Delta T_C = 8.0$; $1.8 \times 8.0 = 14.4$.

Answer with units: +8.0 K and +14.4 °F changes.

Physical interpretation: No offset appears because the zero-point offsets cancel.

Common confusions

Kelvin readings have no ° symbol. Multiplying 20 °C by two does not mean twice the absolute temperature. Celsius is valid for a temperature difference in many energy equations, but not for an absolute-temperature gas-law ratio.

Real-life connection

Device predictions are submitted in Celsius, while a gas-law calculation uses Kelvin. Clearly write “reading” or “change” beside a number before converting it.

Recap

Reading conversions use offsets; change conversions do not. Kelvin and Celsius steps have equal size. Use the scale required by the equation or competition rule.

Review questions

1. Convert 0.0 °C into K.

2. Explain why a 7 °C rise is a 7 K rise.

3. Which operation comes first when converting 77 °F to °C?

Practice

Open WS05. Worksheet questions are separate from the review questions above.

Mastery checklist

☐ I can convert in both directions between the three scales.

☐ I can convert a temperature change without adding an offset.

☐ I can name why Kelvin is needed for absolute-temperature relationships.

U1-C06 – Heat-energy units and numerical conversions

Unit 1 | All competition levels | public-student

Learning objectives

Convert heat-related energy units; distinguish power from energy; use unit cancellation to check a calculation.

Prerequisites

Decimal multiplication/division, ratios, minutes-to-seconds conversion, Unit 1 temperature distinctions.

Learn the idea

Energy can be measured in different units just as a length can be measured in meters or centimeters. The standard SI energy unit is the joule, J. A kilojoule is 1000 J. A number becomes smaller when you rewrite the same energy in a larger unit: 2500 J is 2.5 kJ. The physical energy has not changed.

A small calorie, cal, is another energy unit. In this packet we use the thermochemical calorie: 1 cal=4.184 J exactly. Historical definitions of calorie differ slightly, so a problem should identify the convention when it matters. A kilocalorie is 1000 cal. A food-label Calorie, capital C, commonly means a kilocalorie: 1 Cal=1 kcal=4184 J. Confusing cal with Cal creates a factor-of-1000 error.

The name “calorie” connects to heating water, but an energy unit is not restricted to one material or transfer mechanism. Joules and calories can measure the same transferred energy. A temperature is still a different quantity: there is no universal conversion from joules to degrees Celsius. How much a sample's temperature changes depends on its mass, material and whether a phase change occurs.

Power is the rate at which energy is transferred or converted. One watt is one joule per second. A 5 W transfer delivers 5 J each second. Energy accumulates over time, so E=Pt when power is constant. Use seconds if power is in W and energy is wanted in J. A watt-hour is therefore energy, not power: 1 Wh=3600 J.

A useful unit-cancellation method is to multiply by a fraction equal to one. To convert 3.2 kJ to J, multiply 3.2 kJ by (1000 J)/(1 kJ). The kJ cancels, leaving J. For the opposite conversion, invert the factor. The units show whether you chose the direction correctly before you trust the arithmetic.

The energy emitted or supplied by a lamp is not necessarily the energy absorbed by the water. Some energy misses the device, is reflected away, heats other objects, or leaves to the surroundings. The device unit will build energy balances that include those paths. Never infer the water's energy gain simply by multiplying the bulb's rated electrical power by time and assuming perfect transfer.

For significant figures, conversion constants do not add precision to the measured value. Keep a few extra digits during arithmetic and round the final result to the precision requested by the

question. A full answer includes a number, a unit, and a sentence checking whether its size is reasonable.

Definitions and units

Joule J: energy unit. Kilojoule kJ: 1000 J. Thermochemical calorie cal: 4.184 J. Kilocalorie kcal: 4184 J. Food Calorie Cal: kcal. Watt W: J/s, a power unit. Watt-hour Wh: energy transferred at 1 W for 1 hour.

Equations and their meaning

1 kJ=1000 J; 1 cal=4.184 J; 1 kcal=1000 cal=4184 J; 1 W=1 J/s; E=Pt at constant power; 1 Wh=3600 J. E is energy, P is power, t is elapsed time.

Worked example 1: Calories to joules

Given: Energy transfer is 25.0 cal; use 4.184 J/cal.

Find: Energy in J.

Equation: $E_J=E_{cal}\times 4.184$ J/cal.

Substitution: $E_J=25.0$ cal$\times 4.184$ J/cal.

Arithmetic: $25.0\times 4.184=104.6$.

Answer with units: 105 J to three significant figures.

Physical interpretation: The energy is the same; the numerical value is larger because a joule is smaller than a calorie.

Worked example 2: Energy from constant power

Given: An illustrative transfer provides 6.0 W for 3.0 min.

Find: Energy transferred in J and kJ.

Equation: $E=Pt$; 1 min=60 s.

Substitution: $t=3.0\times 60=180$ s; $E=6.0$ J/s$\times 180$ s.

Arithmetic: $6.0\times 180=1080$ J; $1080/1000=1.080$ kJ.

Answer with units: 1.1×10^3 J, or 1.1 kJ, to two significant figures.

Physical interpretation: This is transferred energy under the stated constant-rate model, not automatically the energy absorbed by a particular water sample.

Worked example 3: A food-label conversion

Given: A label reports 2.00 Cal.

Find: Equivalent energy in kJ.

Equation: 1 Cal=4.184 kJ.

Substitution: E=2.00 Cal×4.184 kJ/Cal.

Arithmetic: 2.00×4.184=8.368.

Answer with units: 8.37 kJ.

Physical interpretation: Capital-C Calories are kilocalories; using 4.184 J per Cal would be 1000 times too small.

Common confusions

W is power; Wh is energy. A calorie is not a temperature. Always distinguish cal, kcal and Cal. Lamp power times time describes supplied energy, while water gain requires a transfer model.

Real-life connection

Power and time help bound possible device energy input, but real heating predictions require measured or modeled absorption and losses. Unit cancellation prevents misleading comparisons between J, kJ and W.

Recap

Convert with ratios and cancel units. Keep power separate from energy. Use seconds with W when calculating J. Energy-to-temperature problems need additional information.

Review questions

1. How many joules are in 0.60 kJ?

2. Why is Wh an energy unit?

3. Why cannot a lamp’s rating alone determine final water temperature?

Practice

Open WS06. Worksheet questions are separate from the review questions above.

Mastery checklist

☐ I can convert J, kJ, cal, kcal and Cal.

☐ I can compute E=Pt with consistent time units.

☐ I can explain why W and °C cannot be substituted for J.

Unit 1 formula and unit packet

public-student | All levels | This is the Unit 1 subset; the full-curriculum packet remains pending.

F01 - Celsius to Kelvin

$T_K = T_C + 273.15$

Units: T_K: K; T_C: °C

Use: An absolute reading, not a temperature change.

Mini example: 20.0 °C → 293.15 K before rounding.

F02 - Kelvin to Celsius

$T_C = T_K - 273.15$

Units: T_K: K; T_C: °C

Use: Convert a Kelvin reading.

Mini example: 300.00 K → 26.85 °C.

F03 - Celsius to Fahrenheit

$T_F = (9/5)T_C + 32$

Units: T_F: °F; T_C: °C

Use: Convert a reading.

Mini example: 5 °C → 41 °F.

F04 - Fahrenheit to Celsius

$T_C = (T_F - 32) \times 5/9$

Units: T_F: °F; T_C: °C

Use: Subtract the offset before rescaling.

Mini example: 59 °F → 15 °C.

F05 - Temperature changes

$\Delta T = T_{final} - T_{initial}$; $\Delta T_K = \Delta T_C$; $\Delta T_F = 1.8\Delta T_C$

Units: ΔT_C: °C change; ΔT_K: K change; ΔT_F: °F change

Use: Never add 273.15 or 32 to a change.

Mini example: A +10 °C rise is +10 K and +18 °F.

F06 - Density

$\rho = m/V$

Units: ρ: g/mL or kg/m³; m: g or kg; V: mL or m³

Use: Match the mass and volume units to the requested density unit.

Mini example: 40 g/50 mL=0.80 g/mL.

F07 - Energy at constant power

$E = Pt$

Units: E: J; P: W (J/s); t: s

Use: Only if the stated transfer power is constant; supplied energy need not all reach the water.

Mini example: 2 W×30 s=60 J.

F08 - Energy conversions

1 kJ=1000 J; 1 cal=4.184 J; 1 kcal=1 Cal=4184 J; 1 Wh=3600 J

Units: cal: thermochemical small calorie; Cal: food Calorie; Wh: watt-hour

Use: Specify calorie convention and capitalize carefully.

Mini example: 0.50 kcal=2.092 kJ before rounding.

Unit 1 glossary

public-student | Full seven-unit glossary pending.

Boundary: The real or imaginary surface separating a chosen system from its surroundings.

Calorie (Cal): A food-label energy unit usually equal to 1 kilocalorie, or 4184 J in this packet.

calorie (cal): The small thermochemical energy unit defined here as 4.184 J.

Closed system: A system that can exchange energy but not matter with its surroundings.

Density: Mass per volume, $\rho = m/V$; an intensive property at specified conditions.

Energy: A conserved physical quantity transferred by heating, work and other routes; measured in joules.

Extensive property: A property that scales with the amount of a system in a controlled same-state comparison.

Heat: Energy transferred because of a temperature difference; a transfer, not stored material.

Homogeneous: Uniform in composition and relevant properties on the scale being considered.

Intensive property: A property that does not simply scale with the amount of a same-state sample.

Internal energy: Microscopic energy associated with a system’s particles and interactions, excluding whole-object motion and gravitational energy.

Isolated system: An ideal system exchanging neither energy nor matter with its surroundings.

Joule (J): The SI unit of energy.

Kelvin (K): The absolute-temperature unit with the same step size as a Celsius degree.

Kilojoule (kJ): 1000 joules.

Net transfer: Transfer in one direction minus transfer in the opposite direction.

Open system: A system that can exchange matter and energy with its surroundings.

Power: Energy transfer or conversion per unit time; measured in watts.

Surroundings: Everything outside the chosen system boundary.

System: The selected matter or region studied in a problem.

Temperature: A thermal-state quantity that determines the direction of spontaneous net heat transfer.

Temperature change: Final temperature minus initial temperature; offsets cancel in scale conversions.

Thermal energy: A classroom term for temperature-related microscopic energy; usage varies, so use internal energy in precise balances.

Thermal equilibrium: Equal-temperature condition giving no net heat transfer between bodies in thermal contact.

Watt (W): One joule per second, a unit of power.

Watt-hour (Wh): An energy unit equal to 3600 J.

Zeroth law: If two systems are each in thermal equilibrium with a third, they are in thermal equilibrium with each other.

Unit 1 review sheet

public-student | All levels | Use after WS01–WS06.

Six ideas to retrieve without notes

1. Temperature is a thermal-state quantity; internal energy belongs to a system; heat is an energy transfer caused by a temperature difference.

2. Choose a boundary before classifying a system. Open exchanges matter and energy; closed exchanges energy but not matter; ideal isolated exchanges neither.

3. Compare portions at the same state: mass, volume and internal energy are extensive; temperature, pressure and density are intensive.

4. Thermal equilibrium means equal temperature and no net heat transfer, not no particle motion. A common reference connects temperature comparisons through the zeroth law.

5. Reading conversions use offsets. Temperature-change conversions do not. Kelvin has no degree symbol.

6. J and kJ measure energy; W measures power. Use seconds in E=Pt when P is in W. Keep cal and food Cal distinct.

A five-step problem routine

1. Name the system and requested quantity.

2. Write the given values with units; say whether temperature is a reading or change.

3. Select a relationship and state any assumption.

4. Substitute consistent units, calculate, and round once at the end.

5. Interpret the sign and size. Does the answer describe energy, power or temperature?

Retrieval tasks

Explain the spoon/bucket comparison. Draw a closed-system energy arrow. Split a sample and name changing properties. Explain a thermometer. Convert readings, changes and energy units.

Error checks before moving on

If you wrote “heat inside the cup,” revisit WS01. If a seal made you say isolated, revisit WS02. If two 20 °C samples became 40 °C, revisit WS03. If equilibrium meant equal energy, revisit WS04. If you added 273.15 to a rise, revisit WS05. If minutes went directly into W×time or Cal meant cal, revisit WS06.

Readiness

Target 16/20 per worksheet and explain corrections. This academy guideline is not tournament scoring.